

AI Usage Note: Quantitative Research Assignment

Candidate Name: Shourya Singh

Project: NIFTY 50 Crash Recovery Quantitative Study

AI Tool Used: Gemini (Google) as a technical writing and research collaborator

1. Overview of AI Integration

AI was utilized strictly as an interactive sounding board and technical assistant. All core analytical decisions, hypothesis design, mathematical frameworks, and interpretations of results were directed, reviewed, and validated by the author to meet strict quantitative finance standards.

2. Specific Areas of Collaboration

- **Code & Engineering:** Assisted in drafting modular Python scripts (yfinance), vectorizing metrics, and avoiding look-ahead bias using next-day Open execution (shift(-1)).
- **Statistical Testing:** Provided syntactic guidance for implementing Welch's T-Tests (scipy.stats) to calculate p-values against market baselines.
- **Document Structuring:** Helped outline the Research Note to maintain a professional, objective tone when reporting a negative result.

3. Human Oversight and Verification

- **Result Validation:** All metrics—including 94 crash events at the -2.0% threshold and the p-value ($p = 0.5193$)—were independently verified via code execution in Google Colab.
- **Analytical Conclusion:** The author upheld the conclusion that the recovery pattern is statistically insignificant (random noise), resisting the temptation of backtest optimization.

4. Reflection

Collaborating with an AI agent expedited the engineering and drafting timeline, allowing full focus to shift toward deeper statistical testing (threshold sensitivity and chronological out-of-sample splits) rather than boilerplate syntax generation.

